

Biometric Authentication Mechanisms

Abstract—Biometric credentials are widely used in modern authentication: Sensors obtain a measurement and output *accept* if it is above a set match threshold or *reject* otherwise. A large body of work improves the accuracy of biometric sensors, but the state of the art uses a heuristic threshold. Other works study *authentication mechanisms*, but those treat biometric credentials as *atomic* (accept/reject, using the heuristic threshold).

We explore optimal authentication mechanisms with biometric credentials. Following prior work, we define a mechanism’s success as the probability it accepts legitimate users and rejects attackers. We formally prove that in general the optimal threshold differs from the common heuristic. Moreover, a different optimum should be used according to the integration with other credentials. We also prove that integrating a biometric credential improves mechanism success even when an atomic credential has arbitrary low fault probabilities and a biometric credential provides only weak separation between users and attackers. We quantitatively demonstrate the improvement with experiments on real public biometric data. Our optimized threshold reduces mechanism failure probability by 30-70% compared to the heuristic threshold when a biometric credential is integrated with other credentials.

These results imply that biometric sensors should allow integrators to set thresholds according to the authentication mechanism.

1. Introduction

Biometric credentials are one of the most prevalent keys in modern authentication. Historically, they were used in signatures and handprints on royal documents and contracts, as they rely on unique physical characteristics of the user made them unforgeable. In the early days of the Internet, biometric credentials were not applicable and therefore *atomic credentials*, like physical devices or passwords, either a perfect match or wrong, were adopted. With the rapid growth of online services, more daily activities has transitioned to digital platforms. This expansion led companies to deploy advanced authentication mechanisms, such as multi-factor authentication (MFA) [1], increasing significantly the number of atomic credentials a user must have or remember to authenticate. However, password vulnerabilities persist: 59% of users in the U.S. reported leaks and 22% frequently forget their passwords, as of 2026 [2]. Consequently, biometric credentials are increasingly adopted, present in most modern user devices, where access to most smartphones relies on FaceID or Fingerprint. The biometric sensors obtain a sample, calculate a similarity score between a sample and

a reference stored in the database, and output accept if the score is above a set match threshold or reject otherwise.

Previous work (§2) has conducted extensive analysis of biometric credentials in isolation, primarily focusing on system accuracy and performance, aiming to reduce the false acceptance (*FAR*) and false rejection (*FRR*) rates [3], [4]. Sarkar et al. [5] introduce the Equal Error Rate (*EER*, where $FAR = FRR$) a heuristic operating point of biometric sensors. Hong et al. [6] propose a fused biometric system in which the final decision is based on the matching scores of both face and fingerprint. While these works widely cover biometric credentials, none of them analyze their integration into an authentication mechanism. Other works proposed authentication mechanisms defined as predicate of credential possession [1], [7], [8], [9], where each credential is treated as atomic. However, such settings are partially applicable to our framework. While we adopt their definition of *mechanism’s success* as the probability it accepts users and rejects attackers, they treat biometric as atomic credentials. This leaves open the question of whether adapting biometric credentials in the presence of other credentials, can form the authentication mechanism to enhance its success.

In this work, we study how biometric credentials influence the authentication mechanism’s success with and without atomic credentials. More precisely, our goal is to optimally use biometric credentials in authentication mechanisms and analyze how the presence of atomic credentials may adapt the biometric credential to enhance mechanism’s success. Fuse accepted models to form the authentication model (§3) built on a set of biometric and atomic credentials: A mechanism interacts with two players: a user and an attacker, trying to identify the user based on the provided credentials by the player. Each credential can be either [9], (1) *safe*: Only available to the user, (2) *lost*: Not available to either player, (3) *leaked*: Available to both players, or (4) *stolen*: Only available to the attacker. Each credential has a probability of being in each of the four states. Biometric credentials are evaluated by a sensor that calculates a confidence score for either player. If the score is above a *match threshold*, the mechanism approves an access to the player, otherwise it denies. This is used to establish two fault probabilities: *false acceptance rate (FAR)*: The probability that the mechanism incorrectly accepts an attacker as a legitimate user, and *false rejection rate (FRR)*: The probability that the mechanism incorrectly rejects a legitimate user.

We start by optimizing a single biometric credential mechanism (§4). We examine distribution functions that approximate a biometric credential score distribution like

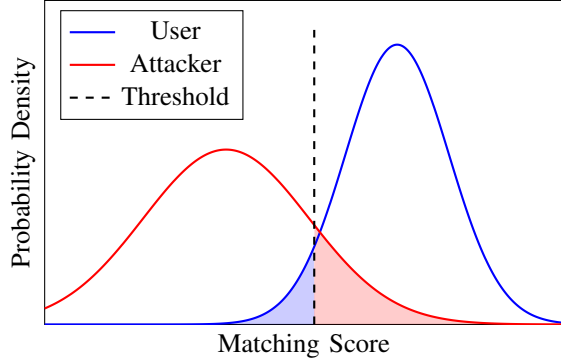


Figure 1: User and Attacker score distributions with FAR and FRR areas appear in red and blue respectively.

shown in Figure 1. We find closed-form expressions of the mechanism’s success for different score distribution functions of a biometric credential and we symbolically calculate the optimal and *EER* thresholds. Our findings show that in general, the optimal threshold differs from the commonly used *EER*.

We examine a more complex case, where a mechanism comprises a biometric and an atomic credential (§5). Such case is common in cloud authentication. For example, Google [10] assume I (1) Possess a trusted device like phone or computer (Atomic credential) and (2) Authenticates via Passkeys [11], e.g, FaceID or Fingerprint. We approximate the user and attacker distribution functions using first and second order approximations, based on the findings of Slobodan et al. [12]. We analyze the mechanism success which reveals that the optimal threshold for a mechanism comprising multiple biometric credentials is not necessarily the same as the optimal threshold if used individually (§5.1). Using numerical optimization, we observe a reduction of over 66% in the failure rate, when comparing the *EER* and the optimal threshold. Moreover, we find close form expressions of a mechanism’s success when atomic and biometric credentials are integrated and symbolically calculate the optimal and *EER* thresholds. These expressions reveal that the mechanism success can be primarily dependant on an atomic credential or the biometric credential, depends on the atomic probability values. For example, when using *AND mechanism*: The user has full availability to all credentials, while the attacker has limited or no availability; The relation shows that high safe values and low leak values of the atomic credential consider a player as a user even when obtaining a sample with low score value, what makes the mechanism’s success mainly dependent on the atomic credential, and vice versa.

Motivated by this observation, we analyze whether the integration of biometric credential with an atomic credential is beneficial in terms of mechanism’s success. We link between the biometric credential to the improvement it has on the mechanism’s success composes an atomic credential and characterize the conditions under which such integration improves success (§6). We prove that when these conditions are not satisfied, it is not worthwhile to integrate, as the atomic

credential mechanism dominates the integrated mechanism. Furthermore, we show that every biometric credential which is *asymptotic separable*: for every score above or below a specific threshold, the error rates converge to zero; a necessary and sufficient condition, provided that the atomic credential has non-zero probabilities for the states of *safe* and either *loss* or *leak*. We also introduce *asymptotic dominance*: where the genuine distribution dominates the attacker distribution by a factor c from every score above a specific threshold, or the attacker dominates the user by a factor c for every score below a specific threshold. This condition is a sufficient condition and shown as it represent a behavior we expect to observe in biometric credentials.

We quantitatively demonstrate our theoretical analysis with experiments using real public biometric data and a standard algorithm for fingerprint verification [13] (§7). We use a dataset of fingerprints from multiple users and evaluate matching scores by comparing samples of one user against those of other users. When different users attempt to authenticate as one another, the biometric sensors succeed in distinguishing and preventing unintended user access. In this setting, we observe high separability between the two players, as the algorithm effectively extracts minutiae and distinguishes between different individuals, resulting in low *EER*. Thus, no significant improvement is observed when comparing the mechanism’s success under Equal Error Rate (*EER*) threshold to the optimal threshold. However, since an attacker does not attempt to authenticate using their own biometric features, the relevant threat model involve attempts to physically impersonate a legitimate user. We conduct a spoofing attack experiment using the LivDet database [14], [15], [16], [17], which includes both genuine fingerprint scans and corresponding spoofing attempts. These spoof samples produce matching scores that closely resemble those of the legitimate user. Integrating the biometric credential into a mechanism comprised of an atomic credential, reduces the mechanism’s failure probability by 30-70% compared to the probability with the *EER* threshold. Furthermore, the results show a 50% reduction in the mechanism’s failure probability compared to a mechanism based solely on an atomic credential, demonstrating the benefit of integrating the biometric credential.

In summary (§8) our contributions are both theoretical and practical: (1) Formalization of the mechanism’s success function composed of biometric and atomic credentials; (2) Symbolic analysis of the optimal threshold for biometric and atomic authentication mechanisms. (3) Characterization of the conditions under which biometric credential integration enhances the mechanism’s success; (4) Proof that any biometric credential, under a specified condition, always enhances the mechanism’s success. (5) Empirical analysis using real biometric data, demonstrating the theoretical analysis. (6) Direct implications to how fingerprint readers should work in practice for immediate security improvements.

2. Related Work

Authentication remains an active and prominent research area, with prior work broadly categorized into two main aspects: credentials and mechanisms. Credentials form the core of the authentication process, commonly manifested as passwords [18], OTPs [19], physical devices [20] or biometric features such as fingerprints, face recognition or iris [21]. An authentication mechanism is a decision making protocol that evaluates a set of credentials provided by a player and output accept or reject [22], [23]. Examples include multi-factor authentication (MFA) [1], an authentication mechanism composes multiple and different credentials' categories treating each one as an atomic credential.

Extensive work has evaluated and improved the latency and accuracy of biometric systems [3], [14], [15], [16], [17], [24], [25]. Prabhakar et al. [4] examine the tradeoff between two key error metrics of a biometric system, namely False Acceptance Rate (*FAR*) and False Rejection Rate (*FRR*). Sarkar et al. [5] introduce the heuristic Equal Error Rate (*EER*), the point where *FAR* and *FRR* are equal, and determine it as an operating point of the biometric system. Hong et al. [6] integrate fingerprint and face recognition to design a fused biometric system that combines the matching scores of both modalities to make a final decision.

Critically, none of these works define an optimizable success criterion for biometric systems. While Hong et al. closely related to our domain, they don't explore the influence of one biometric credential on the other nor do they analyze the integration of biometric credentials along with other credential types in the authentication mechanism. Moreover, Sarkar et al. assume a heuristic operating threshold for the biometric system and do not explore how the presence of additional credentials can influence the optimal matching threshold of the biometric system.

Other work has explored authentication mechanism defined as predicate of credential possession. Maram et al. [7] study interactive authentication mechanisms, where the user and attacker interact with the authentication mechanism. Mouallem et al. [8] compare mechanisms' success and aim to identify mechanisms with maximal success when composed of a large number of homogeneous credentials. Eyal [9] analyzes mechanisms modeled as Boolean functions of the credentials. However, these works are partially applicable to our setting. While we adopt their mechanism's success definition and the way they define the credential possession states, they model biometric credentials as atomic credentials. This leaves open the question of whether the integration of biometric credentials along with other credentials, can form the authentication mechanisms to improve its success, which is the focus of our work.

3. Authentication Model

We formalize the authentication problem specifying the system components (§3.1). We furtherly define each credential's type and their characteristics (§3.2), the authentication

mechanism (§3.3) and the relation between different mechanisms.

3.1. System Components

The system comprises three main components: a set of credentials $\{c_1, c_2, \dots, c_n\}$, an authentication mechanism M and two players: a user U and an attacker A . All three are finite and deterministic automata.

To authenticate into the user assets, each player take advantage of his private information to satisfy the authentication mechanism. The private information of a player is represented by the credentials available to him. Credentials can be represented in different forms like data (e.g, passwords), physical devices (e.g, security keys, smart-phones and laptops) or biometric features (e.g, fingerprint or FaceID). Each credential has a public part, known to the mechanism and a secret part, known only to the player. For example, when trying to access a website using a username and a passwords, a public part is the hash stored in the database and its secret part is the password and username themselves.

3.2. Credential

Definition 1. (Credential) A credential c is a tuple $c = (c^P, c^S)$ that can be either *atomic* or *biometric*. c^P represents the public part of the credential and c^S represents the secret part.

The system uses a set of credentials $\{c_1, c_2, \dots, c_n\}$. Each credential c_i can be either *atomic* or *biometric* and can belong to one of the states: *Safe*, *Loss*, *Leak* or *Stolen*. We define the probability associated with each state as follows: P_{safe} , P_{loss} , P_{leak} and P_{stolen} . An attacker A and a user U have different availability to each credential. For example, a password may be available to the user but not to the attacker, while a leaked password may be available to both. An *availability vector* represents the availability of all credentials to a user U or an attacker A . Denote by σ^U, σ^A the availability vector of the user and the attacker respectively. A scenario σ is the vector of states of each credential. Denote by σ_i the state of credential c_i in scenario σ . For example, if $\sigma_i = \text{leak}$ then $\sigma_i^U = \text{True}$ and $\sigma_i^A = \text{True}$.

An *atomic credential* is either available or not to a player. The system checks a perfect match of the atomic credential between an input of a player and a reference. If a user U provides an atomic credential and $\sigma_i = \text{leak}$ then both $\sigma_i^U = \text{True}$ and $\sigma_i^A = \text{True}$ means the system grant access to both players. A *biometric credential* is evaluated by a biometric sensor. Denote by s the *matching score* that quantifies the similarity between an input obtained by the sensor and the reference stored. Let t denote the *match threshold*. If the score is above the threshold, the system approves an access to the player, otherwise it denies [4], [21]. The matching score produced by the biometric sensor characterizes the players distributions: The genuine distribution $f_U[s]$ in the bounds $[u_1, u_2]$, is generated when a

sample obtained by the sensor belongs to the real user. The impostor distribution $f_A[s]$ in the bounds $[a_1, a_2]$, is generated when a sample obtained by the sensor belongs to an attacker. These distributions establish two error metrics: *FAR* and *FRR* both are functions of the match threshold t . If a user U provides a biometric credential and $\sigma_i = \text{safe}$ then $\sigma_i^U = \text{True}$ and $\sigma_i^A = \text{False}$. We use the *FAR* and *FRR* to define the biometric credential availability states:

$$P_{\text{loss}} = FRR(t) \cdot (1 - FAR(t))$$

$$P_{\text{leak}} = FAR(t) \cdot (1 - FRR(t))$$

$$P_{\text{stolen}} = FRR(t) \cdot FAR(t)$$

$$P_{\text{safe}} = 1 - P_{\text{loss}} - P_{\text{leak}} - P_{\text{stolen}} = (1 - FAR(t)) \cdot (1 - FRR(t))$$

Where *FAR* and *FRR* are given by:

$$FAR(t) = \int_t^{a_2} f_A[s] ds$$

$$FRR(t) = \int_{u_1}^t f_U[s] ds$$

3.3. Mechanism

Definition 2. (Authentication Mechanism) An authentication mechanism M is a finite deterministic function that takes as input a set of credentials $\{c_1, c_2, \dots, c_n\}$ and outputs a decision of either *accept* or *reject*.

An authentication mechanism M interact with two players: a user U and an attacker A . Both players, try to satisfy the mechanism with their own set of credentials $\{c_1, c_2, \dots, c_n\}$ in order to reach the asset. The decision of the authentication mechanism is based on the mechanism's logic.

Definition 3. (Mechanism Logic) A mechanism logic L is the decision rule that maps credential's availability σ_i to a final outcome *accept* or *reject* under certain logical conditions.

For example, in an *AND mechanism*, the mechanism grants access to a user if and only if all credentials are available to the user and at least one of them not available to the attacker. In an *OR mechanism*, the mechanism grants access to a user if at least one credential is available to the user and none are available to the attacker. We are now ready to define the mechanism's success.

Definition 4. (Mechanism Success) Mechanism success is the probability that the authentication mechanism M successfully accepts a user U and rejects an attacker A given their set of credentials $\{c_1, c_2, \dots, c_n\}$ and the mechanism logic L .

For a mechanism M comprised of a single credential c_1 , the mechanism success is simply the probability that the credential is neither theft, loss or leaked for every mechanism logic L . i.e.,

$$P_{\text{success}}(M^{\text{single}}) = P_{\text{safe}} \quad (1)$$

For a mechanism M comprised of two credentials c_1 and c_2 , the mechanism logic L determines the relation between the two credentials and thus the mechanism's success. There are two scenarios determines the mechanism success:

a. *AND*: The user has full availability to all credentials, while the attacker has limited or no availability. For successful authentication, we need both credentials to be either in safe state, or one in safe state and the other in leak state. If one of the credentials is either loss or theft, the owner cannot satisfy the mechanism and thus the mechanism fails.

$$P_{\text{Success}}(M^{\text{AND}}) = P_{\text{safe}}^1 \cdot P_{\text{safe}}^2 + P_{\text{safe}}^1 \cdot P_{\text{leak}}^2 + P_{\text{leak}}^1 \cdot P_{\text{safe}}^2 \quad (2)$$

b. *OR*: The user has availability to at least one credential, while the attacker has availability to neither of them. For successful authentication, we need both credential to be either in safe state, or one in safe state and the other in loss state. If one of the credentials is either leak or theft, the attacker can satisfy the mechanism and thus the mechanism fails.

$$P_{\text{Success}}(M^{\text{OR}}) = P_{\text{safe}}^1 \cdot P_{\text{safe}}^2 + P_{\text{safe}}^1 \cdot P_{\text{loss}}^2 + P_{\text{leak}}^1 \cdot P_{\text{loss}}^2 \quad (3)$$

4. Single Biometric Credential

Until now, it has been common practice [5] to select the operating point at the Equal Error Rate (*EER*), where $FAR = FRR$. Having defined different mechanisms success, we proceed to analyze a mechanism comprises first (§4.1) and second (§4.2) order approximations for a single biometric credential. We show that the optimal threshold differs from the *EER* and explore the relation between the mechanism success and the *EER*.

4.1. Uniformly Distributed Credential

As a first-order approximation, we model the biometric credential as a uniform distribution. Denote by u_1, u_2 the user PDFs' bounds and a_1, a_2 are the attacker PDFs' bounds. We defined the probability distribution functions for the user and the attacker, f_U and f_A as follows:

$$f_U(s) = \frac{1}{u_2 - u_1} \quad \text{for } u_1 \leq s \leq u_2$$

$$f_A(s) = \frac{1}{a_2 - a_1} \quad \text{for } a_1 \leq s \leq a_2$$

The *FAR* (t) and *FRR* (t) are given by:

$$FAR(t) = \int_t^{a_2} \frac{1}{a_2 - a_1} ds = \begin{cases} 1 & \text{if } t \leq a_1 \\ \frac{a_2 - t}{a_2 - a_1} & \text{if } a_1 < t \leq a_2 \\ 0 & \text{if } t > a_2 \end{cases}$$

$$FRR(t) = \int_{u_1}^t \frac{1}{u_2 - u_1} ds = \begin{cases} 0 & \text{if } t \leq u_1 \\ \frac{t - u_1}{u_2 - u_1} & \text{if } u_1 < t \leq u_2 \\ 1 & \text{if } t > u_2 \end{cases}$$

Denote:

$$\alpha(t) = \frac{t - u_1}{u_2 - u_1}, \quad \beta(t) = \frac{a_2 - t}{a_2 - a_1}.$$

We expect the user distribution to be characterized by higher values of the matching score s than the attacker. Otherwise, the attacker would be more likely to produce scores that lead to successful authentication, rendering the biometric credential ineffective. Thus, we assume that $a_2 < u_2$. We analyze three different cases:

(1) $a_2 < u_1$: In this case, the attacker and user distributions are completely separated. Any threshold t chosen within the separation region results in perfect mechanism success.

(2) $u_1 < a_1 < a_2 < u_2$: The attacker distribution is fully contained by the user distribution. The probability of success is given by Equation 1. By setting the $FAR(t)$ and $FRR(t)$ into Equation 1, we calculate the probability of success.

$$P_{\text{success}}(t) = \begin{cases} 0, & t \leq u_1 \\ 0, & u_1 < t \leq a_1 \\ 1 - \alpha(t) - \beta(t) + \alpha(t) \cdot \beta(t), & u_1 < t \leq a_2 \\ 1 - \alpha(t), & a_2 < t \leq u_2 \\ 0, & t > u_2 \end{cases}$$

(3) $a_1 < u_1 < a_2 < u_2$: The user and attacker distributions partially overlap. Figure 2 illustrates this case where the user and attacker PDF's are symmetrically and asymmetrically distributed both as function of the matching score s .

By setting the $FAR(t)$ and $FRR(t)$ into Equation 1, the probability of success is given by:

$$P_{\text{success}}(t) = \begin{cases} 0, & t \leq a_1 \\ 1 - \beta(t), & a_1 < t \leq u_1 \\ 1 - \alpha(t) - \beta(t) + \alpha(t) \cdot \beta(t), & u_1 < t \leq a_2 \\ 1 - \alpha(t), & a_2 < t \leq u_2 \\ 0, & t > u_2 \end{cases}$$

We analyze the probability of success and find the optimal threshold that maximizes the mechanism's success in each interval. We note that the third and fourth regions in both second and third cases are the same. Thus, we can analyze them together to find the optimal threshold in each region. The success function is linearly increasing in the second region of the third case, thus the maximum is achieved at the right bound of the region where $t = u_1$. The success function is linearly decreasing in the fourth region in both second and third cases, thus the optimal threshold in this region is at the left bound of the region, which is $T = a_2$. Since $P_{\text{success}}(T)$ is a continuous function, the global maximum is expected to achieve in the third region in both cases. Then, we analyze the third region and find the global maximum and optimal threshold:

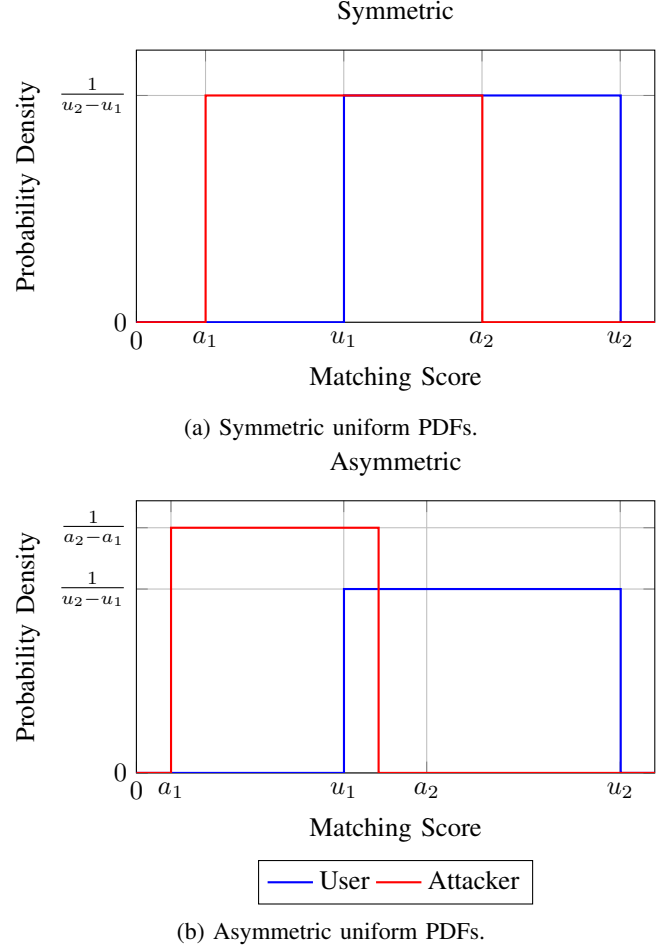


Figure 2: Symmetric and asymmetric uniform PDFs.

We derive the success function in the third region and set it to zero to find the horizontal tangent points:

$$\frac{d}{dt} P_{\text{success}}(t) = 0.$$

The horizontal tangent point is:

$$t = \frac{a_1 + u_2}{2}$$

$$\frac{d}{dt} P_{\text{success}}(t) > 0, \text{ For every } t \in (u_1, \frac{a_1 + u_2}{2})$$

$$\frac{d}{dt} P_{\text{success}}(t) < 0, \text{ For every } t \in (\frac{a_1 + u_2}{2}, a_2)$$

Thus, by the First Derivative Test the maximum point is at $t_{\text{Max}} = \frac{a_1 + u_2}{2}$.

$$P_{\text{success}}(t_{\text{Max}}) = \frac{(a_1 - u_2)^2}{4(a_1 - a_2)(u_1 - u_2)}$$

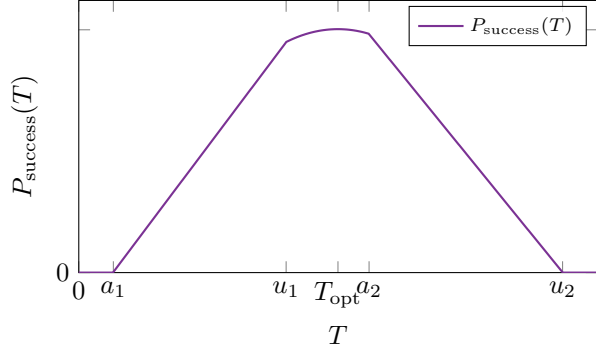


Figure 3: Success probability as a function of the threshold.

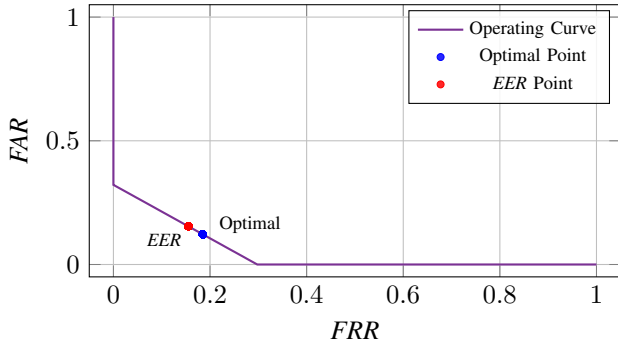


Figure 4: *FAR* vs. *FRR* operating curve with optimal and *EER* points.

We calculate the *EER* point and show it is not equal to the optimal point:

$$t_{EER} = \frac{a_1 \cdot u_1 - a_2 \cdot u_2}{a_1 - a_2 + u_1 - u_2} \neq \frac{a_1 + u_2}{2} = t_{Max}$$

The maximum point represents the optimal threshold of the mechanism. The optimal point indicates that increasing the offset between the two uniform distributions is likely to improve the mechanism's success. Moreover, it suggests an asymmetry between the user's and the attacker's distribution intervals. Figure 3 illustrates the probability of success for a single biometric credential mechanism as a function of threshold t . We show how the operating point influence the error metrics *FAR* and *FRR* in Figure 4. The *FAR* and *FRR* at the optimal point are given by:

$$FAR(t_{opt}) = \frac{2a_2 - a_1 - u_2}{2(a_2 - a_1)}$$

$$FRR(t_{opt}) = \frac{a_1 + u_2 - 2u_1}{2(u_2 - u_1)}$$

Indicating that symmetry between the user and attacker distributions implies that the optimal operating point coincides with the *EER*. We illustrate the improvement of the mechanism's success between the optimal point and the *EER* point when shifting u_1 to lower values of the score s and shifting a_1 to higher values of the score s in Figure 5.

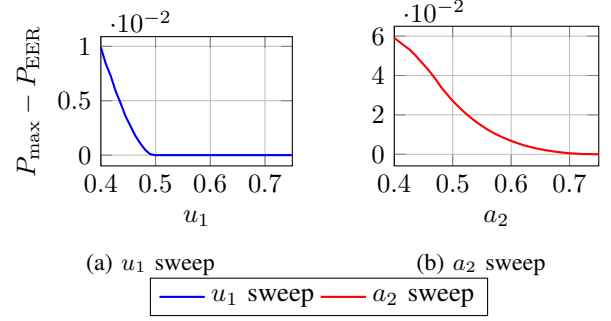


Figure 5: Difference between the optimal success probability and the success probability at *EER* under parameter sweeps.

4.2. Quadratic Distributed Credential

Our goal is to model credentials whose distributions closely resemble those observed in real measurement data. To this end, we analyze a quadratic probability distribution function, which provides a second order approximation of the biometric credential that can also be seen in real fingerprint measurement data (§7). This choice is also consistent with the findings of Slobodan et al. [12] on Eigenfinger and Eigenpalm feature distributions. Accordingly, we define two probability density functions (PDFs) for the user and the attacker, denoted by f_U and f_A , respectively.

$$f_U(s) = \begin{cases} a \cdot s^2 + b \cdot s + c & \alpha_1 \leq s \leq \alpha_2 \\ 0 & s > \alpha_2 \vee s < \alpha_1 \end{cases}$$

$$f_A(s) = \begin{cases} d \cdot s^2 + e \cdot s + f & \beta_1 \leq s \leq \beta_2 \\ 0 & s > \beta_2 \vee s < \beta_1 \end{cases}$$

Here, a, b, c and d, e, f denote the quadratic coefficients of the user and attacker distributions, respectively. The parameters α_1, α_2 are the roots of the user's PDF, while β_1, β_2 are the roots of the attacker's PDF. Similarly, to the analysis of the uniform distribution, the x axis represents the matching score, such that higher values of score will be correspondant to the reference template stored in the database X_U . If $\beta_2 < \alpha_1$, the two distributions do not overlap, and the problem becomes trivial. Conversely, if $\alpha_1 > \beta_1$ and $\beta_2 < \alpha_2$, the attacker's distribution is fully contained within the user's, resulting in a non-separable scenario. We therefore focus on the intermediate and most meaningful configuration, where

$$\beta_1 \leq \alpha_1 \leq \beta_2 \leq \alpha_2,$$

as illustrated in Figure 6.

To reduce the degrees of freedom of the quadratic functions while maintaining a symbolic analysis, we express the coefficients directly in terms of their roots. Under the additional assumption that the integral of each PDF over its roots interval equals 1, the coefficients become fully determined. Accordingly, we define the coefficients as follows.

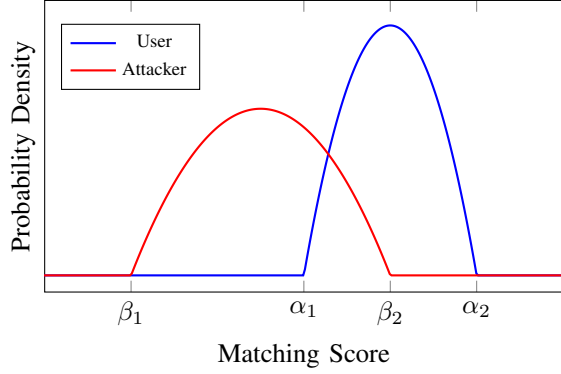


Figure 6: Quadratic user and attacker distributions.

$$\alpha_1 \cdot \alpha_2 = \frac{c}{a} \rightarrow a = \frac{c}{\alpha_1 \cdot \alpha_2}$$

$$\alpha_1 + \alpha_2 = -\frac{b}{a} \rightarrow b = -\frac{c \cdot (\alpha_1 + \alpha_2)}{\alpha_1 \cdot \alpha_2}$$

$$\beta_1 \cdot \beta_2 = \frac{f}{d} \rightarrow d = \frac{f}{\beta_1 \cdot \beta_2}$$

$$\beta_1 + \beta_2 = -\frac{e}{d} \rightarrow e = -\frac{f \cdot (\beta_1 + \beta_2)}{\beta_1 \cdot \beta_2}$$

$$\int_{\alpha_1}^{\alpha_2} (a \cdot x^2 + b \cdot x + c) dx = 1 \Rightarrow c = \frac{6 \cdot \alpha_1 \cdot \alpha_2}{(\alpha_1 - \alpha_2)^3}$$

$$\int_{\beta_1}^{\beta_2} (d \cdot x^2 + e \cdot x + f) dx = 1 \Rightarrow f = \frac{6 \cdot \beta_1 \cdot \beta_2}{(\beta_1 - \beta_2)^3}$$

Therefore, we can write the quadratic PDFs as:

$$f_U(s) = \begin{cases} \frac{6}{(\alpha_1 - \alpha_2)^3} s^2 - \frac{6(\alpha_1 + \alpha_2)}{(\alpha_1 - \alpha_2)^3} s + \frac{6\alpha_1\alpha_2}{(\alpha_1 - \alpha_2)^3} & \alpha_1 \leq s \leq \alpha_2 \\ 0, & \text{otherwise} \end{cases}$$

$$f_A(s) = \begin{cases} \frac{6}{(\beta_1 - \beta_2)^3} s^2 - \frac{6(\beta_1 + \beta_2)}{(\beta_1 - \beta_2)^3} s + \frac{6\beta_1\beta_2}{(\beta_1 - \beta_2)^3} & \beta_1 \leq s \leq \beta_2 \\ 0, & \text{otherwise} \end{cases}$$

We will find *FAR* and *FRR* as a function of the threshold t and define the probability of success based on a single credential mechanism as shown in (§4.1). Define:

$$A(t) = \frac{2(\beta_2^3 - t^3) - 3(\beta_1 + \beta_2)(\beta_2^2 - t^2) + 6\beta_1\beta_2(\beta_2 - t)}{(\beta_1 - \beta_2)^3},$$

$$U(t) = \frac{2(t^3 - \alpha_1^3) - 3(\alpha_1 + \alpha_2)(t^2 - \alpha_1^2) + 6\alpha_1\alpha_2(t - \alpha_1)}{(\alpha_1 - \alpha_2)^3}.$$

Thus,

$$FAR(t) = \begin{cases} 1, & t \leq \beta_1, \\ A(t), & \beta_1 < t \leq \beta_2, \\ 0, & t > \beta_2, \end{cases}$$

$$FRR(t) = \begin{cases} 0, & t \leq \alpha_1, \\ U(t), & \alpha_1 < t \leq \alpha_2, \\ 1, & t > \alpha_2. \end{cases}$$

$$G(t) = -\frac{(t - \alpha_2)^2(t - \beta_1)^2(2t - 3\alpha_1 + \alpha_2)(2t + \beta_1 - 3\beta_2)}{(\alpha_1 - \alpha_2)^3(\beta_1 - \beta_2)^3}.$$

$$P_{\text{success}}(t) = \begin{cases} 0, & t < \beta_1 \vee t > \alpha_2, \\ \frac{(t - \beta_1)^2(2t + \beta_1 - 3\beta_2)}{(\beta_1 - \beta_2)^3}, & \beta_1 \leq t \leq \alpha_1, \\ G(t), & \alpha_1 \leq t \leq \beta_2, \\ -\frac{(t - \alpha_2)^2(2t - 3\alpha_1 + \alpha_2)}{(\alpha_1 - \alpha_2)^3}, & \beta_2 \leq t \leq \alpha_2. \end{cases}$$

We differentiate $P_{\text{success}}(t)$ and compute its critical points in every interval. By comparing the stationary values obtained in all valid regions, we identify the global maximum, which corresponds to the optimal threshold t_{opt} .

Before writing the piecewise derivative, we define the polynomial coefficients and the following function:

$$q = 4,$$

$$m = -(5\alpha_1 + \alpha_2 + \beta_1 + 5\beta_2),$$

$$n = \alpha_1(3\alpha_2 + \beta_1 + 6\beta_2) - \alpha_2^2 + \alpha_2\beta_2 - \beta_1^2 + 3\beta_1\beta_2,$$

$$\ell = \alpha_1(\beta_1^2 - 3\beta_2(\alpha_2 + \beta_1)) + \alpha_2^2\beta_2.$$

$$R(t) = -\frac{6(t - \alpha_2)(t - \beta_1)(qt^3 + mt^2 + nt + \ell)}{(\alpha_1 - \alpha_2)^3(\beta_1 - \beta_2)^3}.$$

With these coefficients and function $R(t)$, the derivative of $P_{\text{success}}(t)$ is given by the following piecewise expression:

$$\frac{dP_{\text{success}}}{dt} = \begin{cases} 0, & t < \beta_1 \vee t > \alpha_2, \\ \frac{6(t - \beta_1)(t - \beta_2)}{(\beta_1 - \beta_2)^3}, & \beta_1 \leq t \leq \alpha_1, \\ R(t), & \alpha_1 \leq t \leq \beta_2, \\ -\frac{6(t - \alpha_1)(t - \alpha_2)}{(\alpha_1 - \alpha_2)^3}, & \beta_2 \leq t \leq \alpha_2. \end{cases}$$

In the second interval, the derivative has a single critical point at $t = \beta_1$. Since the derivative is strictly positive for all $t > \beta_1$, this point corresponds to a local minimum rather than a maximum. Similarly, in the fourth interval, the only critical point occurs at $t = \alpha_2$, and the derivative is strictly negative for all $t < \alpha_2$. Therefore, this interval also contains no maximum either.

We therefore turn to the third interval. In this region, the derivative is the product of a cubic polynomial and two linear factors. The linear factors vanish at $t = \beta_1$ and $t = \alpha_2$, both of which lie outside the third interval. Consequently, the existence of critical points within this interval depends solely on the cubic term. Solving this cubic, reveals exactly one real root with the remaining two roots being complex. The closed form of the cubic solution can not be simplified further more than a complex expression. Thus, we denote the solution as t_{cubic} .

By examining the sign of the derivative on both sides of the critical point t_{cubic} , we verify that it constitutes a global

maximum of $P_{\text{safe}}(t)$. We denote this maximizing point by t_{opt} .

To demonstrate that this optimal threshold differs from the Equal Error Rate (*EER*) threshold, we computed the *EER* point separately as follows:

$$t_{\text{EER}} = \frac{\alpha_1\beta_1 - \alpha_2\beta_2}{\alpha_1 - \alpha_2 + \beta_1 - \beta_2} \neq t_{\text{opt}}$$

We illustrate several user and attacker distributions and present their corresponding probability of success curves, expressed as functions of the threshold t in Figure 7 and Figure 8 respectively.

Moreover, we illustrate the *FAR-FRR* curves for each pair of user and attacker distributions, and mark both the optimal point and the *EER* point, as shown in Figure 9.

To quantify the effect of the variance on the deviation between the optimal operating point t_{opt} and the equal error rate (*EER*) point t_{EER} , we numerically computed this gap as a function of the attacker's variance. Decreasing the variance of the attacker achieved by reducing the distance between its roots, corresponds to reduced separability between the two distributions. The simulations reveal that as the variance increases, the gap $|t_{\text{opt}} - t_{\text{EER}}|$ grows accordingly, until the operating threshold moves into a regime where it effectively minimizes only one of the error rates, either *FAR* or *FRR*, driving the other to zero. This phenomenon is shown in Figure 10. These findings indicate that higher attacker variance enlarges the potential success advantage gained by operating at t_{opt} rather than at the *EER* point.

5. Two Credentials

For a mechanism M comprised of two independently and identically distributed credentials, derived from either the same or different PDF's, we defined the integrated system success as the probability of users successfully defend their assets under two scenarios:

a. *AND*: The user has full availability to all credentials, while the attacker has limited or no availability. For successful authentication, we need both credentials to be either in safe state, or one in safe state and the other in leak state. If one of the credentials is either loss or theft, the owner cannot satisfy the mechanism and thus the mechanism fails.

$$P_{\text{Success}}(M^{\text{AND}}) = P_{\text{safe}}^1 \cdot P_{\text{safe}}^2 + P_{\text{safe}}^1 \cdot P_{\text{leak}}^2 + P_{\text{safe}}^2 \cdot P_{\text{leak}}^1 \quad (4)$$

b. *OR*: The user has availability to at least one credential, while the attacker has no availability neither of them. For success, we need both credential to be either in safe state, or one in safe state and the other in loss state. If one of the credentials is either leak or theft, the attacker can satisfy the mechanism and thus the mechanism fails.

$$P_{\text{Success}}(M^{\text{OR}}) = P_{\text{safe}}^1 \cdot P_{\text{safe}}^2 + P_{\text{safe}}^1 \cdot P_{\text{loss}}^2 + P_{\text{safe}}^2 \cdot P_{\text{loss}}^1 \quad (5)$$

For each mechanism type, we analyze the case of two independently and identically distributed credentials in 2 cases:

- A mechanism comprised of 2 biometric credentials.
- A mechanism comprised of 1 atomic credential and 1 biometric credential.

We derive the optimal operating point for both *AND* and *OR* mechanisms by maximizing the respective success probabilities as function of the thresholds of each credential.

5.1. 2 Biometric Credentials

To analyze the case of a mechanism M comprised of two biometric credentials, we conduct a numerical study over several probability distribution functions. For each distribution pair, we derive the optimal operating point for both *AND* and *OR* mechanisms by maximizing the respective success probabilities with respect to the thresholds of each credential, using a LBFGS optimization algorithm. We additionally compare the optimal operating point to the *EER* point, thereby characterizing the conditions under which the *AND* or *OR* mechanisms yields superior performance. We started by analyzing the case of Gaussian distributions. We defined the user and attacker distributions as $P_U(s) = \mathcal{N}(\mu_u, \sigma_u^2)$ and $P_A(s) = \mathcal{N}(\mu_a, \sigma_a^2)$ respectively. We set for the first credential $\mu_u = 0.8$, $\sigma_u = 0.12$, $\mu_a = 0.3$ and $\sigma_a = 1$ and for the second credential $\mu_u = 1.0$, $\sigma_u = 0.2$, $\mu_a = 0.4$ and $\sigma_a = 0.18$ as shown in Figure 11.

The optimal operating point for both mechanisms' types differ from the *EER* point, yielding higher success probabilities than the *EER* points. Moreover, we find that the optimal points of both *AND* and *OR* mechanisms are not necessarily the same as the optimal threshold if used individually. We furtherly find that such integration of a biometric credential to a mechanism already comprised of a biometric credential, improves the mechanism success. Hence, integrating two biometric credentials into a mechanism improves its success over using them individually as shown in Figure 12.

We illustrate the success probability of both *AND* and *OR* mechanisms. We fix a credential to its optimum found out using the LBFGS algorithm and show how the success probability varies as a function of the thresholds of the other credential. The results are shown in Figure 13. In conclusion, all numerical results are summarized in Table 1.

From Table 1, we observe that the optimal operating point yields a higher success probability than the *EER* point by between 5% to 7%.

5.2. 1 Atomic and 1 Biometric Credential

We start by analyzing the case of an *AND* mechanism comprised of 1 atomic credential and 1 biometric credential. Denote the state of credential by s_i , i.e., $i \in (\text{Atomic}, \text{Biometric})$, we define the atomic credential with a constant value of each probabilities' states as: $P_{\text{safe}}^1, P_{\text{loss}}^1, P_{\text{leak}}^1, P_{\text{theft}}^1$ corresponding respectively to each one of the states: safe, loss, leak and theft. The biometric credential is defined by its PDFs for the user and the attacker, P_U and P_A respectively. We model the first order approximation of the biometric credential using uniform

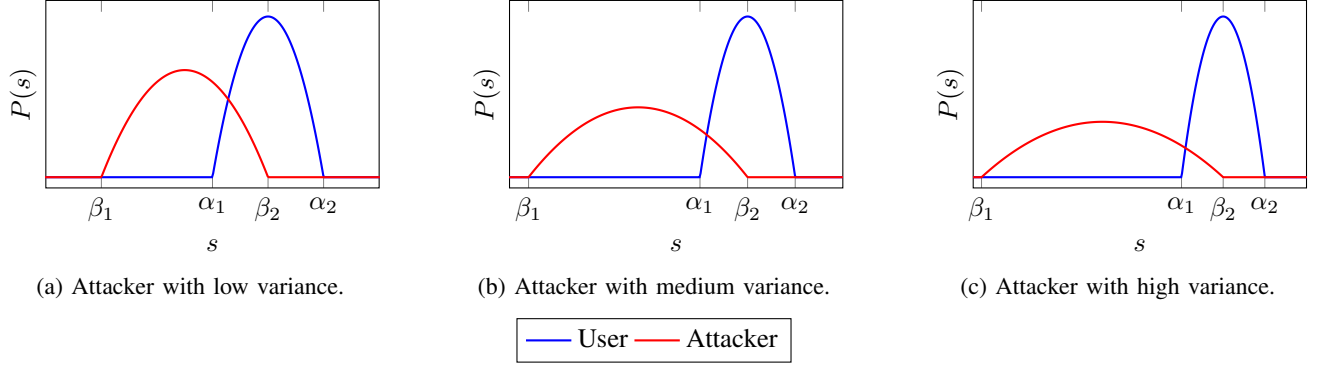


Figure 7: Quadratic user and attacker distributions.

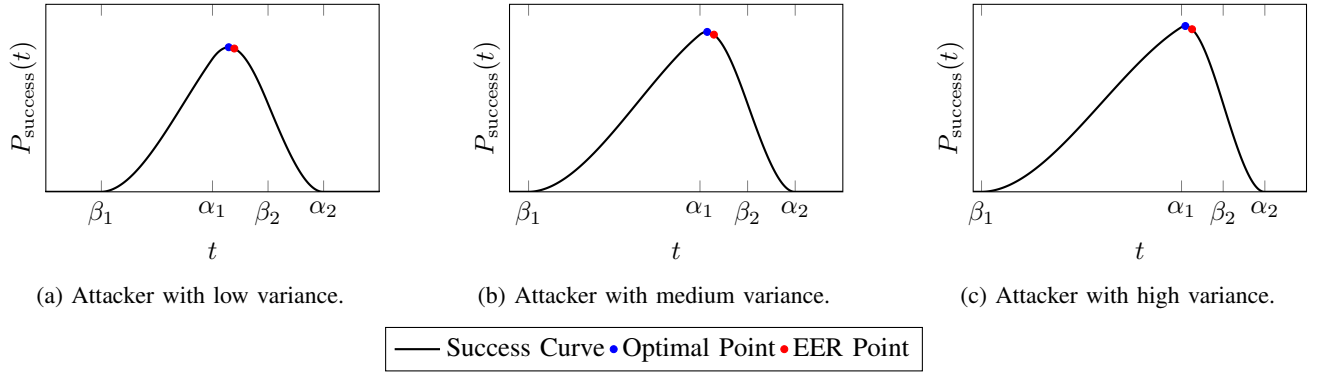


Figure 8: Success probability for three quadratic attacker distributions.

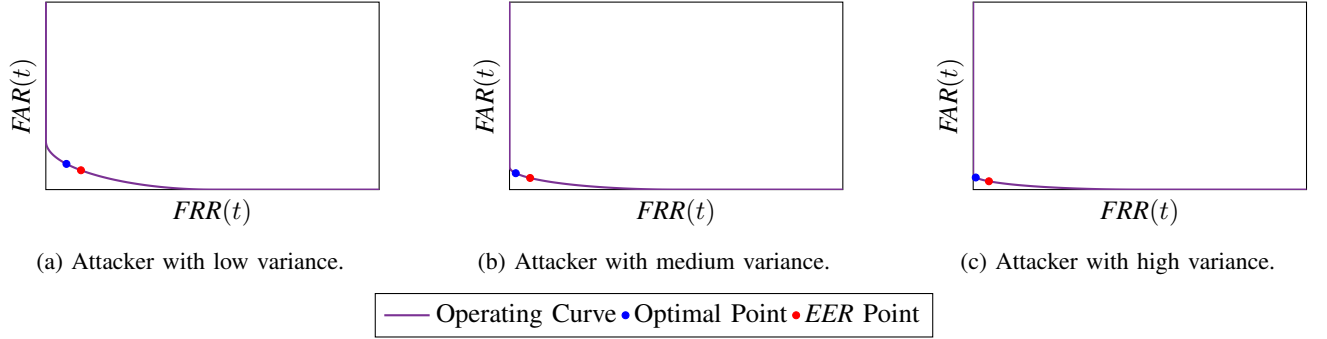


Figure 9: FAR vs. FRR operating curves for different attacker distributions.

distribution between the bounds of u_1, u_2 corresponding to the user distribution and a_1, a_2 corresponds to the attacker, as seen at (§4.1). We derive the probability of the biometric credential as follows:

$$P_{safe}^2 = 1 - FAR(t) - FRR(t) + FAR(t) \cdot FRR(t),$$

$$P_{leak}^2 = FAR(t) (1 - FRR(t)).$$

To analyze the probability of success for the *AND* mechanism, we set the above probabilities into Equation 4. Thus, the probability of success is a function of the threshold t and divided into 5 intervals. Denote:

$$\alpha(t) = \frac{u_2 - t}{u_2 - u_1}, \quad \beta(t) = \frac{a_2 - t}{a_2 - a_1}.$$

$$L(t) = \alpha(t) [(P_{safe}^1 + P_{leak}^1) - P_{leak}^1 \cdot \beta(t)].$$

$$P_{success}(t) = \begin{cases} P_{safe}^1, & t \leq a_1 \\ P_{safe}^1 + P_{leak}^1(1 - \beta(t)), & a_1 < t \leq u_1 \\ L(t), & u_1 < t \leq a_2 \\ \alpha(t) \cdot (P_{safe}^1 + P_{leak}^1), & a_2 < t \leq u_2 \\ 0, & t > u_2. \end{cases}$$

We analyze the success probability $P_{success}(t)$ by partitioning the threshold domain into five intervals. Each interval corresponds to a distinct relative positioning of the threshold with respect to the attacker and user distributions.

Distribution	$P_{\text{AND}}(t_{\text{EER}})$	$P_{\text{OR}}(t_{\text{EER}})$	$P_{\text{AND}}(t_1^{\text{safe}})$	$P_{\text{OR}}(t_2^{\text{safe}})$	$P_{\text{AND}}(t_{\text{opt}})$	$P_{\text{OR}}(t_{\text{opt}})$
Gaussian	0.910	0.911	0.911	0.911	0.964	0.962
Uniform	0.955	0.955	0.916	0.999	1.000	1.000
Parabolic	0.920	0.921	0.919	0.922	0.987	0.985

TABLE 1: Mechanism success under Gaussian, Uniform and Parabolic score distributions at three reference operating points: EER-pair $(t_1^{\text{EER}}, t_2^{\text{EER}})$, individual optimum pair $(t_1^{\text{safe}}, t_2^{\text{safe}})$, and joint optima (AND at $(t_1^*, t_2^*)_{\text{AND}}$ and OR at $(t_1^*, t_2^*)_{\text{OR}}$)

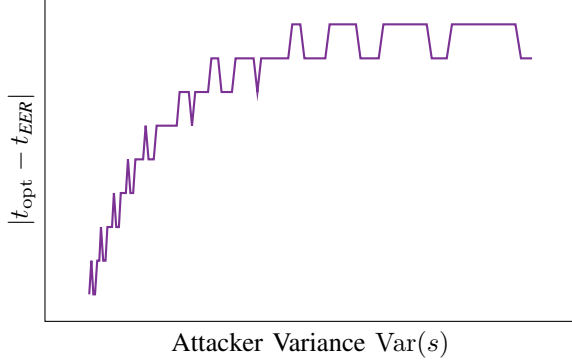


Figure 10: Gap between optimal and EER thresholds as a function of attacker's variance.

The first interval, $t \leq a_1$, corresponds to the case in which the threshold is lower than the minimum value of the attacker distribution. In this regime, the FAR is equal to 1, while the FRR is equal to 0. Consequently, the safe probability of the biometric credential is zero, and the success probability converges to P_{safe}^1 . Since P_{safe}^1 is constant in this interval, no interior maximum exists.

The second interval, $a_1 < t \leq u_1$, corresponds to the case in which the threshold lies between the minimum value of the attacker distribution and the minimum value of the user distribution. In this region, the FAR decreases linearly as a function of t , while the FRR remains equal to 0. As a result, the probabilities of the safe state and the leak state of the biometric credential are $1 - \text{FAR}(t)$ and $\text{FAR}(t)$, respectively. In this interval, $P_{\text{success}}(t)$ is a strictly increasing function, as its derivative is a constant positive value, and therefore the maximum is attained at the boundary point: $t_{\text{opt}2} = u_1$.

The third interval, $u_1 < t \leq a_2$, corresponds to the case in which the threshold lies between the minimum value of the user distribution and the maximum value of the attacker distribution. In this region, both FAR and FRR vary linearly with t , resulting in a success probability that is quadratic as a function of t . Differentiating $P_{\text{success}}(t)$ yields a single critical point inside the interval,

$$t_{\text{opt}3} = \frac{u_2 + a_2 - (a_2 - a_1) \frac{P_{\text{safe}}^1 + P_{\text{leak}}^1}{P_{\text{leak}}^1}}{2}$$

Since this expression is valid only for the interval $u_1 < t \leq a_2$, the effective maximizer within this interval is

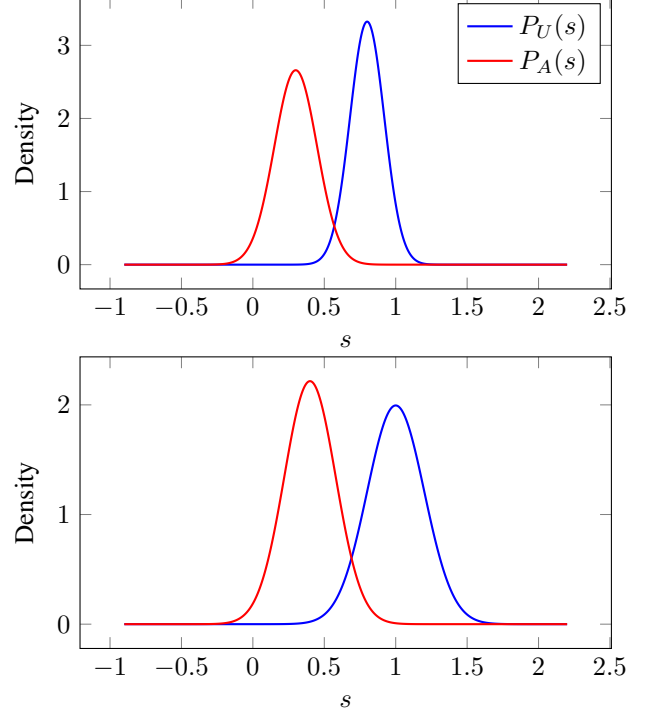


Figure 11: Gaussian user and attacker distributions.

obtained by clipping the optimum to the interval boundaries, as follows:

$$t_{\text{opt},3}^{(\text{interval})} = \min(a_2, \max(u_1, t_{\text{opt},3})).$$

In particular, if $t_{\text{opt},3} < u_1$, the maximum over this interval is attained at $t = u_1$, whereas if $t_{\text{opt},3} > a_2$, the maximum is attained at $t = a_2$. In all configurations consisting of one atomic and such biometric credential, ingertating the mechanism with the biometric credential monotonically enhances its success probability. Evaluating the success probability at the user threshold $t = u_1$ yields

$$P_{\text{success}}(u_1) = P_{\text{safe}}^1 + P_{\text{leak}}^1 \left(1 - \frac{a_2 - u_1}{a_2 - a_1}\right).$$

Since $u_1 > a_1$, it follows that

$$P_{\text{success}}(u_1) > P_{\text{safe}}^1,$$

Demonstrating that the integration of the biometric credential strictly improves the success of the mechanism compared to using the atomic credential alone.

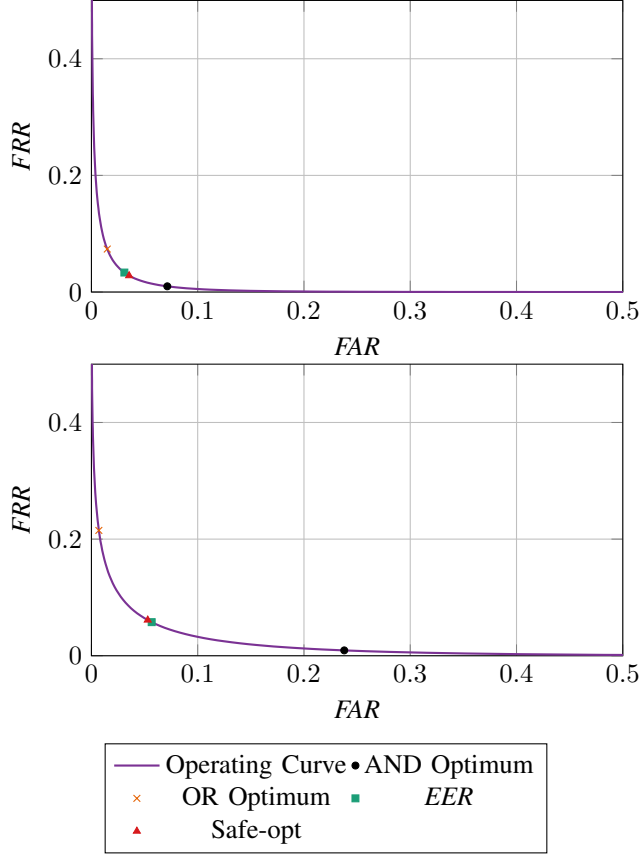


Figure 12: FAR vs. FRR operating curves for two biometric credentials.

By examining the optimal point in this interval under *AND* mechanism, we can deduce that as the variance of the attacker distribution increases, the optimal threshold t_{opt3} shifts toward the lower boundary of the interval, u_1 . Moreover, for an atomic credential with high safe values P_{safe}^1 , and low leak values P_{leak}^1 , the optimal threshold shifts toward the lower boundary as well. Thus, in such case, the success of the mechanism becomes mainly dependent on the atomic credential. Conversely, an atomic credential with high leak values P_{leak}^1 and low safe values P_{safe}^1 causes the optimal threshold to shift toward the upper boundary of the interval, a_2 . In this scenario, the success of the mechanism relies more heavily on the biometric credential.

The fourth interval, $a_2 < t \leq u_2$, corresponds to the case in which the threshold exceeds the maximum value of the attacker distribution. In this regime, the FAR is equal to 0, while the FRR increases linearly with t . Consequently, $P_{success}(t)$ is a strictly decreasing function in this interval, and the maximum is attained at the boundary point

$$t_{opt4} = a_2.$$

Finally, the fifth interval, $t > u_2$, corresponds to the case in which the threshold exceeds the maximum value of the user and the attacker distributions, leaving the success probability equal to 0.

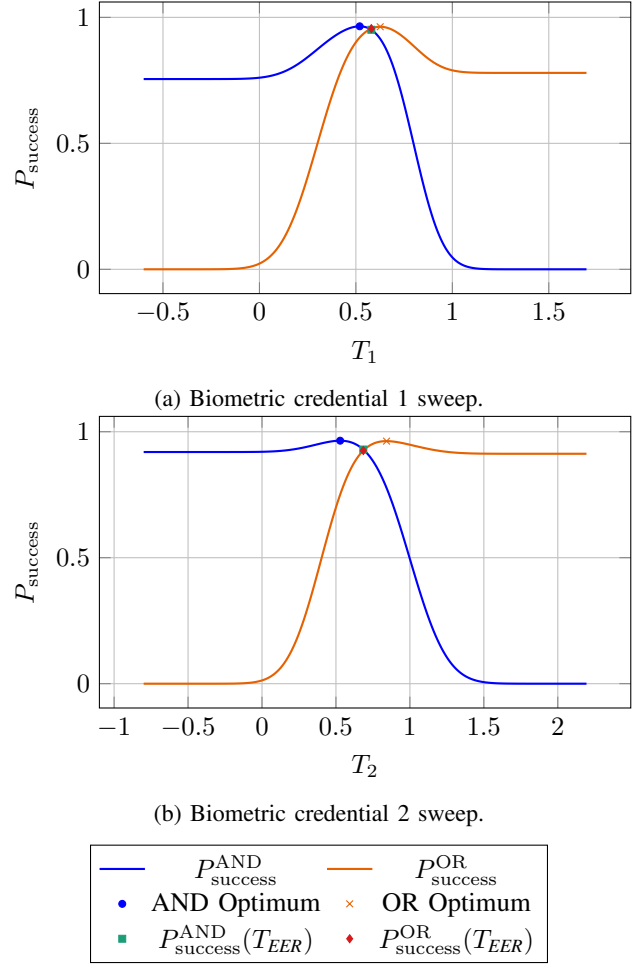


Figure 13: Success probability as a function of the threshold for two biometric credentials.

To determine the global optimal operating point, we compare all stationary points and boundary maxima obtained in the valid intervals. The optimal threshold is achieved in the third interval, where the attacker and user distributions overlap and the success function attains an interior maximum. Substituting $t_{opt} = t_{opt3}$ into $P_{success}(t)$ yields

$$P_{success}(t_{opt}) = \frac{(-a_1(P_{safe}^1 + P_{leak}^1) + a_2P_{safe}^1 + P_{leak}^1 u_2)^2}{4P_{leak}^1(a_2 - a_1)(u_2 - u_1)}.$$

We have done similar analysis for the *OR* mechanism. We defined the credentials' probabilities of the mechanism as follows:

$$\begin{aligned} P_{safe1} &= 1 - FAR(t) - FRR(t) + FAR(t) \cdot FRR(t), \\ P_{safe2} &= k_1, \\ P_{loss1} &= FRR(t) (1 - FAR(t)), \\ P_{loss2} &= k_2. \end{aligned}$$

Therefore, the probability of success for the *OR* mechanism is defined as:

$$P_{\text{success}}(t) = \begin{cases} 0, & t \leq a_1 \\ \frac{t - a_1}{a_2 - a_1} (k_1 + k_2), & a_1 < t \leq u_1 \\ \frac{t - a_1}{a_2 - a_1} \left[k_1 + k_2 - k_2 \frac{t - u_1}{u_2 - u_1} \right], & u_1 < t \leq a_2 \\ k_1 + k_2 - k_2 \frac{t - u_1}{u_2 - u_1}, & a_2 < t \leq u_2 \\ k_1, & t > u_2. \end{cases}$$

By differentiating every interval and identifying the critical points, we found that the optimal threshold that leads to the maximum probability of success lies in the third interval where

$u_1 < t \leq a_2$ and is given by:

$$t_{\text{opt}} = \frac{a_1 k_2 + u_2 (k_1 + k_2) - k_1 u_1}{2k_2}$$

And the maximum probability of success at this point is:

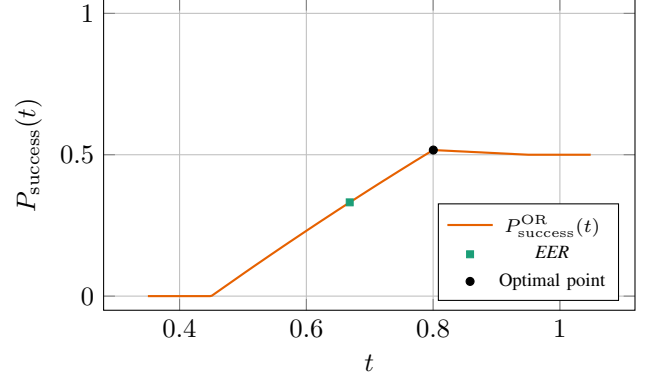
$$P_{\text{success}}(t_{\text{opt}}) = \frac{(a_1 k_2 - u_2 (k_1 + k_2) + k_1 u_1)^2}{4k_2 (a_2 - a_1) (u_2 - u_1)}$$

For both mechanisms, we observe that the optimal operating point t_{opt} differs from the equal error rate (*EER*) point t_{EER} . As the *EER* depends on the PDF's type solely, we calculate the *EER* point and show both mechanism's success in Figure 14

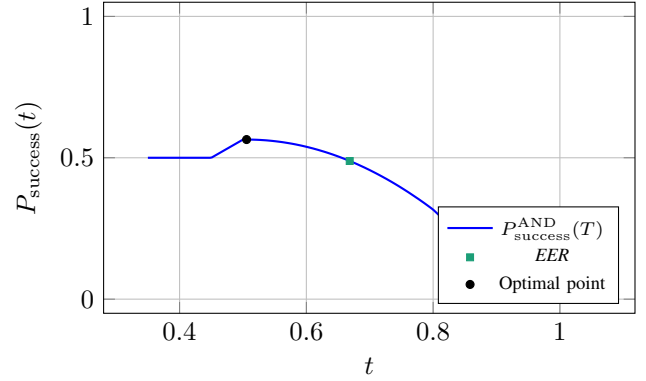
$$t_{\text{EER}} = \frac{a_1 u_1 - a_2 u_2}{a_1 - a_2 + u_1 - u_2}$$

6. Characterizing Success Improvement

We will now characterize the conditions under which integrating a biometric credential into a mechanism composed of an atomic credential can enhance its success. First, we show that for any mechanism whose atomic credential characterized by non-zero probabilities P_{safe} and P_{theft} only, such integration cannot improve mechanism's success (§6.1). Next, we present a necessary and sufficient condition establishing that, for any mechanism composed of an atomic credential with non zero probabilities P_{safe} , P_{leak} or P_{loss} , integrating a biometric credential improves mechanism's success for both finite and infinite distributions (§6.2). Finally, we derive a more common criterion that provides a sufficient condition under which a biometric credential improves the mechanism's success composed of an atomic credential with non zero probabilities P_{safe} , P_{leak} or P_{loss} (§6.3).



(a) OR mechanism.



(b) AND mechanism.

Figure 14: Success probability as a function of the threshold for OR and AND mechanisms composed of an atomic and a biometric credential.

6.1. No Success Gain in the Safe/Theft Case

Proposition 1. For any mechanism M composed of an atomic credential with non-zero probabilities P_{safe} and P_{theft} only, integrating the mechanism with a biometric credential does not enhance its mechanism's success.

Proof. Consider a mechanism M composed of an atomic credential with non-zero probabilities P_{safe} and P_{theft} only. In this case, integrating the mechanism with a biometric credential lead to success probability described by Equation 4 and Equation 5. Since $P_{\text{leak}} = P_{\text{loss}} = 0$, the success probability in both mechanism cases reduces to:

$$P_{\text{success}}(t) = P_{\text{safe}}^1 \cdot P_{\text{safe}}^2(t)$$

Since $P_{\text{safe}}^2(t) \leq 1$ for every value of t , it follows that $P_{\text{success}}(t) \leq P_{\text{safe}}^1$, and thus integrating a biometric credential does not enhance the success of the mechanism. $\square$

6.2. Sufficient And Necessary Condition for Integrating Biometric Credentials

First, we define the notion of *asymptotic separability* of a biometric credential. Then, we will show that its a neces-

sary and sufficient condition for enhancing the mechanism's success for both finite and infinite distributions.

Definition 5. A biometric credential with user distribution function $P_u(s)$ and attacker distribution function $P_a(s)$ is said to be *asymptotically separable* if

$$\inf_t \min \left\{ \frac{FAR(t)}{1 - FRR(t)}, \frac{FRR(t)}{1 - FAR(t)} \right\} = 0,$$

where $FAR(t) = \int_t^\infty P_a(s)ds$ and $FRR(t) = \int_{-\infty}^t P_u(s)ds$.

Proposition 2. For any mechanism M composed of an atomic credential with non-zero probabilities P_{safe} and P_{leak} or P_{loss} , integrating the mechanism with a biometric credential whose user and attacker distributions are asymptotically separable, enhances the success of the mechanism.

Proof. The success of a mechanism composed of an atomic credential with non-zero probabilities P_{safe} and P_{leak} or P_{loss} is given by P_{safe}^1 . A biometric credential is characterized by 4 probability states as function of t : $P_{safe}^2(t)$, $P_{loss}^2(t)$, $P_{leak}^2(t)$, $P_{theft}^2(t)$. We can describe each one of these probabilities as a function of False acceptance rate (FAR) and False rejection rate (FRR) as follows:

$$\begin{aligned} P_{safe}^2(t) &= 1 - FAR(t) - FRR(t) + FAR(t) \cdot FRR(t), \\ P_{loss}^2(t) &= FRR(t)(1 - FAR(t)), \\ P_{leak}^2(t) &= FAR(t)(1 - FRR(t)), \\ P_{theft}^2(t) &= FAR(t) \cdot FRR(t). \end{aligned}$$

The success probability of the integrated mechanism comprised of 2 credentials is described by Equation 4 and Equation 5. Thus, by placing the biometric credential in each configuration we can express the success probability of the integrated mechanism as a function of the operating threshold t :

$$P_{success}^{AND}(t) = (1 - FRR(t)) \left(P_{safe}^1 + P_{leak}^1(1 - FAR(t)) \right). \quad (6)$$

$$P_{success}^{OR}(t) = (1 - FAR(t)) \left(P_{safe}^1 + P_{loss}^1(1 - FRR(t)) \right). \quad (7)$$

We need to show that there exists an operating threshold t such that either $P_{success}^{AND}(t) > P_{safe}^1$ or $P_{success}^{OR}(t) > P_{safe}^1$. By algebraic manipulation of Equation 6 we can demand $P_{success}^{AND}(t) > P_{safe}^1$ if and only if:

$$\frac{FRR(t)}{1 - FAR(t)} < \frac{P_{leak}^1}{P_{safe}^1 + P_{leak}^1 \cdot (1 - FAR(t))}$$

Since $P_{leak}^1 \cdot (1 - FAR(t)) \leq P_{leak}^1$, a sufficient condition for the above inequality is:

$$\frac{FRR(t)}{1 - FAR(t)} < \frac{P_{leak}^1}{P_{safe}^1 + P_{leak}^1} \quad (8)$$

Similarly, we can deduce that $P_{success}^{OR}(t) > P_{safe}^1$ if and only if:

$$\frac{FAR(t)}{1 - FRR(t)} < \frac{P_{loss}^1}{P_{safe}^1 + P_{loss}^1 \cdot (1 - FRR(t))}$$

Since $P_{loss}^1 \cdot (1 - FRR(t)) \leq P_{loss}^1$, a sufficient condition for the above inequality is:

$$\frac{FAR(t)}{1 - FRR(t)} < \frac{P_{loss}^1}{P_{safe}^1 + P_{loss}^1} \quad (9)$$

By the definition of asymptotic separability, there exists an operating threshold t such that:

$$\inf_t \min \left\{ \frac{FAR(t)}{1 - FRR(t)}, \frac{FRR(t)}{1 - FAR(t)} \right\} = 0.$$

Equivalently, $\forall \varepsilon > 0, \exists t$ such that either $\frac{FAR(t)}{1 - FRR(t)} < \varepsilon$ or $\frac{FRR(t)}{1 - FAR(t)} < \varepsilon$. Thus, we can choose ε that equals the right hand side of either Equation 8 or Equation 9 and by definition we know that a t satisfies the inequality exists. More precisely, if

$$\inf_t \frac{FAR(t)}{1 - FRR(t)} = 0,$$

then the OR mechanism composition is preferable, whereas if

$$\inf_t \frac{FRR(t)}{1 - FAR} = 0,$$

then the AND mechanism composition is preferable. Therefore, integrating an asymptotically separable biometric credential enhances the success of the mechanism. $\square$

Proposition 3. Suppose there $\exists t$ such that the integrated mechanism M composed of a biometric credential and an atomic credential with non zero probabilities P_{safe} and P_{leak} or P_{loss} achieves a strictly higher success probability than an atomic credential mechanism, i.e.,

$$P_{success}^{integrated}(t) > P_{success}^{atomic}.$$

Then necessarily, the biometric credential is asymptotically separable.

Proof. We prove by contradiction. Assume for the sake of contradiction, that the biometric credential is not asymptotically separable. Then $\exists \varepsilon > 0$ such that $\forall t$ both $\frac{FAR(t)}{1 - FRR(t)} \geq \varepsilon$ and $\frac{FRR(t)}{1 - FAR(t)} \geq \varepsilon$. Since it's given that the integrated mechanism M achieves a strictly higher success probability than an atomic credential mechanism, then there exists a threshold t such that either $P_{success}^{AND}(t) > P_{safe}^1$ or $P_{success}^{OR}(t) > P_{safe}^1$. Thus, either Equation 8 or Equation 9 holds for every value of $P_{safe}^1, P_{leak}^1, P_{loss}^1$ in the range of $(0, 1)$. In particular, we can choose $P_{safe}^1, P_{leak}^1, P_{loss}^1$ such that either $\frac{P_{leak}^1}{P_{safe}^1 + P_{leak}^1} < \varepsilon$ or $\frac{P_{loss}^1}{P_{safe}^1 + P_{loss}^1} < \varepsilon$ since the equation holds $\forall \varepsilon > 0$, but both $\frac{FAR(t)}{1 - FRR(t)} \geq \varepsilon$ and $\frac{FRR(t)}{1 - FAR(t)} \geq \varepsilon$ derive by the sake of contradiction assumption. Thus, we can deduce that both $\frac{FRR(t)}{1 - FAR(t)} \geq \frac{P_{leak}^1}{P_{safe}^1 + P_{leak}^1}$ and $\frac{FAR(t)}{1 - FRR(t)} \geq \frac{P_{loss}^1}{P_{safe}^1 + P_{loss}^1}$, which contradicts both Equation 8 and Equation 9 under the strictly higher success probability assumption. Therefore, the biometric credential must be asymptotically separable. $\square$

We show that *asymptotic separability* of the user and attacker distributions is a sufficient and necessary condition for enhancing the success of a mechanism composed of an atomic credential with non zero probabilities $P_{\text{safe}}, P_{\text{leak}}$ or P_{loss} by integrating it with a biometric credential.

6.3. Common Sufficient Condition

Definition 6. A biometric credential with user distribution function $P_u(s)$ and attacker distribution function $P_a(s)$ is said to be *asymptotically dominated* if:

$$\forall c > 0 \exists t \text{ s.t. : } \forall s \geq t : p_a(s) \leq c p_u(s)$$

or

$$\forall c > 0 \exists t \text{ s.t. : } \forall s \leq t : p_u(s) \leq c p_a(s)$$

We will show that *asymptotically dominancy* of user and attacker distribution functions is a sufficient condition for enhancing an atomic credential mechanism's success.

Proposition 4. For any mechanism M composed of an atomic credential and success function $P_{\text{success}}^{\text{atomic}}$, integrating the mechanism with an asymptotically dominated biometric credential improves its achievable success. That is,

$$P_{\text{success}}^{\text{integrated}}(t) > P_{\text{success}}^{\text{atomic}}.$$

Proof. An atomic credential is characterized by fixed probability states $P_{\text{safe}}^1, P_{\text{loss}}^1, P_{\text{leak}}^1, P_{\text{theft}}^1$. Thus the success probability of a mechanism composed solely of this atomic credential is constant and given by:

$$P_{\text{success}}^{\text{atomic}} = P_{\text{safe}}^1$$

A biometric credential is characterized by 4 probability states that depend on the operating threshold t : $P_{\text{safe}}^2(t), P_{\text{loss}}^2(t), P_{\text{leak}}^2(t), P_{\text{theft}}^2(t)$. We can describe each one of these probabilities as a function of False acceptance rate (FAR) and False rejection rate (FRR) as follows:

$$\begin{aligned} P_{\text{safe}}^2(t) &= 1 - \text{FAR}(t) - \text{FRR}(t) + \text{FAR}(t) \cdot \text{FRR}(t), \\ P_{\text{loss}}^2(t) &= \text{FRR}(t)(1 - \text{FAR}(t)), \\ P_{\text{leak}}^2(t) &= \text{FAR}(t)(1 - \text{FRR}(t)), \\ P_{\text{theft}}^2(t) &= \text{FAR}(t) \cdot \text{FRR}(t). \end{aligned}$$

The success probability of the integrated mechanism comprised of 2 credentials is described by Equation 4 and Equation 5. Thus, we can express the success probability of the integrated mechanism as a function of the operating threshold t as shown in Equation 6 and Equation 7. We previously observed that a sufficient condition for improvement of mechanism's success is described by Equation 8 and Equation 9. Thus, we need to show that there exists an operating threshold t such that either:

$$\frac{\text{FRR}(t)}{1 - \text{FAR}(t)} < \frac{P_{\text{leak}}^1}{P_{\text{safe}}^1 + P_{\text{leak}}^1}$$

or

$$\frac{\text{FAR}(t)}{1 - \text{FRR}(t)} < \frac{P_{\text{loss}}^1}{P_{\text{safe}}^1 + P_{\text{loss}}^1}$$

We expand left hand side of Equation 8.

$$\frac{\text{FRR}(t)}{1 - \text{FAR}(t)} = \frac{\int_{-\infty}^t P_u(s) ds}{\int_{-\infty}^t P_a(s) ds}$$

By Definition 6, since the biometric credential is asymptotically dominated, it follows that there exists an operating threshold t such that for every $s \leq t$, $P_u(s) \leq c \cdot P_a(s)$. Therefore:

$$\frac{\int_{-\infty}^t P_u(s) ds}{\int_{-\infty}^t P_a(s) ds} \leq \frac{\int_{-\infty}^t c \cdot P_a(s) ds}{\int_{-\infty}^t P_a(s) ds} = c$$

This is true $\forall c > 0$, thus we can choose c such that satisfies Equation 8.

We expand left hand side of Equation 9.

$$\frac{\text{FAR}(t)}{1 - \text{FRR}(t)} = \frac{\int_t^{\infty} P_a(s) ds}{\int_t^{\infty} P_u(s) ds}$$

By Definition 6, since the biometric credential is asymptotically dominated, it follows that there exists an operating threshold t such that for every $s \geq t$, $P_a(s) \leq c \cdot P_u(s)$. Therefore:

$$\frac{\int_t^{\infty} P_a(s) ds}{\int_t^{\infty} P_u(s) ds} \leq \frac{\int_t^{\infty} c \cdot P_u(s) ds}{\int_t^{\infty} P_u(s) ds} = c, \forall c > 0$$

We showed that $\forall c > 0$ either $\frac{\text{FAR}(t)}{1 - \text{FRR}(t)} < c$ or $\frac{\text{FRR}(t)}{1 - \text{FAR}(t)} < c$. Therefore, integrating an asymptotically dominated biometric credential with an atomic credential improves mechanism success. $\square$

This proof is not trivial, as the integration of another atomic credential to an atomic credential mechanism, does not always enhance its success. For example, if the mechanism is comprised of 1 atomic credential with high safe values and low leak values, adding an atomic credential with low safe values and high leak values would decrease the overall mechanism's success. Thus, Proposition 4 shows that by using an *asymptotically dominated* biometric credential, we can always improve the success of all mechanism's composed of an atomic credential. Moreover, *asymptotically dominancy* condition supports practical scenarios, as we expect that for very low values of score s the attacker distribution dominates the user, vice versa for very high values of s . We supports this claim in with real public biometric data in Section 7.

7. Measurements

We now present experiment measurements results based on fingerprints taken from FVC2002 database [26] and FVC2004 database [27]. We used a standard fingerprint verification algorithm *SourceAFIS* created by Vazan [13] for scoring fingerprints out of their images based on minutiae triplets. We chose to use fingerprints, as it is a widely

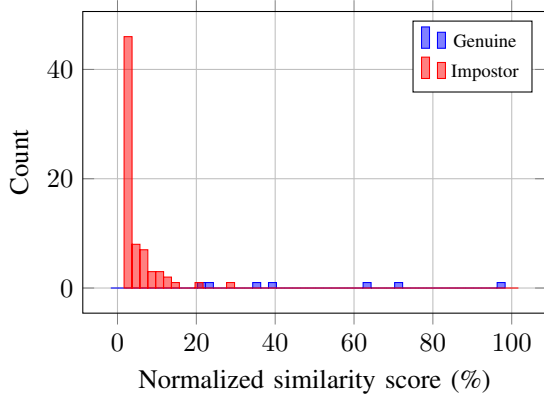


Figure 15: Score histogram for genuine and impostor comparisons.

common and used example of a biometric credential in authentication mechanisms. For a test set, we took a template out of *FVC2002* database 1 of user 103 and compare them against the whole database of both *FVC2002* and *FVC2004*. A genuine will be considered the template of user 103 against its other templates while impostor considered as all the other combinations. For scoring methodology, we define the final score as the maximum similarity across all 5 user templates. For example, if the matching scores between a sample and the 5 probes are [0.1, 0.3, 0.6, 0.9, 0.4], then the resulting score is 0.9.

We maintain same approach as Jain et al. [3] and Cappelli et al. [24] and first built histogram of the genuine and impostor combinations as shown in Figure 15. Then, we turn the histogram into a smoothed probability distribution function. We find the False rejection rate and False acceptance rate and calculate the Equal Error Rate (*EER*) as shown in Figure 16.

The most important outcome from this approach was to achieve similar *EER* of around 2% as achieved from Maio et al. [25], to objectively make a comparison with state-of-the-art algorithms. We calculate the probability success function as given in Equation 1 of a mechanism composed of a biometric credential and show that the optimal operating point is different from the *EER* point and achieves higher success probability as shown in Figure 17. Finally, we illustrate the probability success function of a mechanism composed of an atomic credential with $P_{\text{safe}} = 0.9$ and $P_{\text{leak}} = 0.04$, $P_{\text{loss}} = 0.04$ and integrate it with the biometric credential.

We show that the optimal operating point of the integrated mechanism is different from the *EER* point and achieves higher success probability by around 3% compared to the *EER* point, as the *EER* itself is only 1.8%. In addition, we can observe that the optimal point in both mechanisms achieves higher success probability than both the atomic credential mechanisms and biometric credential mechanism when tested individually as shown in Figure 17. Although there is an improvement in the success probability of the integrated mechanism, we can observe the improvement is

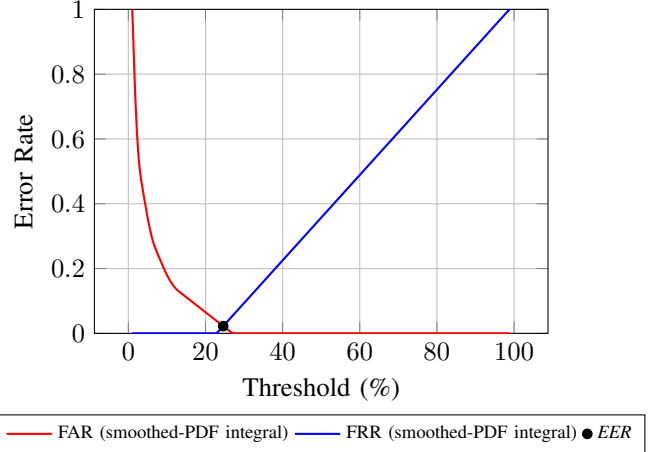
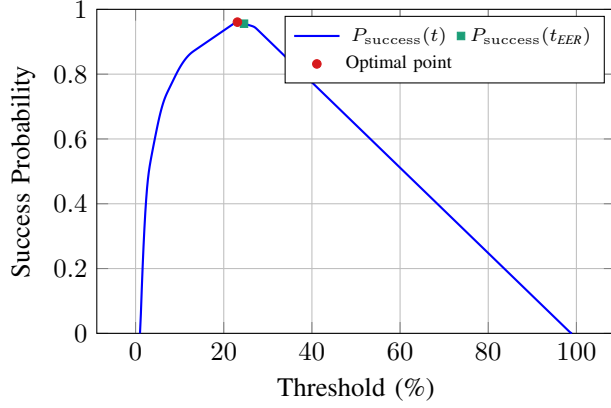


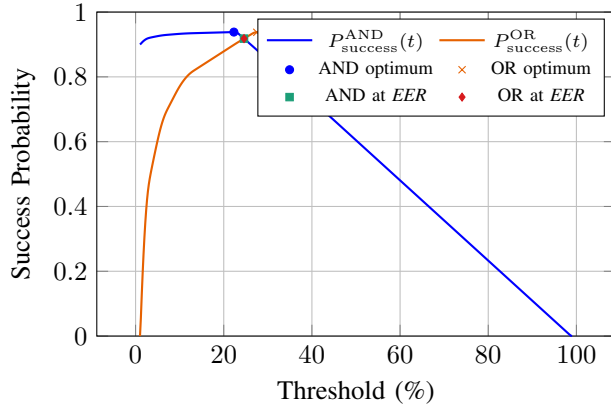
Figure 16: *FAR* and *FRR* as a function of the decision threshold, with the Equal Error Rate (*EER*) point.

not as significant as the one observed in the theoretical analysis. This can be explained by the fact that the *EER* point of the biometric credential is already very low, indicating high separability between a user and an attacker and thus, there is no much room for improvement. In that scenario, we compared several templates of user 103 against templates of different users. However, our main approach in this paper was to increase the mechanism's success in scenarios where an impostor will try to spoof the user templates and authenticate into the user account. Hence, we expect the impostor to use spoofing tools, producing samples resemble with score distribution function that much more similar to the genuine distribution function, which might lead to increased *EER* point.

To support this claim, we used the LivDet2015 database [14], [15], [16], [17], which contains a large-scale database of live and spoof fingerprint samples. We follow the same methodology as before and used the SourceAFIS algorithm [13]. As probe templates, we selected six Digital Persona scanner images of user 002_0. Each probe was matched against 14 additional images of the same user to generate genuine scores. For each target image, we define the final score methodology as the maximum similarity across 6 probe templates. For example, if the matching scores between a target image and the six probes are [0.3, 0.5, 0.4, 0.8, 0.9, 0.7], then the resulting score is 0.9. To simulate a spoofing attack, we used fake fingerprint images of the same user (002_0) produced by Gelatine, Woodglue, Latex and Ecoflex and match them against the probe templates to obtain the impostor score distribution. The same maximum-score aggregation was applied in this case as well. We present the resulting histogram in Figure 18. We then compute the *EER* point and plot the *FAR-FRR* curve, as shown in Figure 19. As expected, the *EER* increased significantly to 17.8%. Finally, we evaluated the success probability for both a single biometric credential and for a mechanism composed of an atomic credential and a biometric credential, as shown in Figure 20. We



(a) Single biometric credential mechanism's success.



(b) biometric credential integrated with an atomic credential into AND and OR mechanisms.

Figure 17: Success probability as a function of the threshold. The AND and OR mechanisms achieve success probabilities of 0.93 and 0.94 at the *EER* point, and 0.96 and 0.95 at their respective optimal operating points.

can observe that the optimal operating point reducing the failure probability of an AND mechanism by 60% and Or mechanism by 44% compared to the *EER* point. Moreover, we reduced the failure probability by 50% when integrating into AND mechanism compared to an atomic credential mechanism with $P_{\text{safe}} = 0.8$, $P_{\text{loss}} = 0.04$, $P_{\text{leak}} = 0.15$, $P_{\text{theft}} = 0.01$ probabilities.

8. Conclusion

We formalize the authentication mechanism model consisting, for the first time, of both biometric and atomic credential mechanisms. We analyze the success of each mechanism type and present a new operating point that achieves higher success probability by modelling each credential with four success probability states. We show every mechanism comprised of atomic credentials can be enhanced by integrating it with asymptotically separable biometric credential. Our analysis shows significant reduction of the failure probability under spoofing attempts, which became

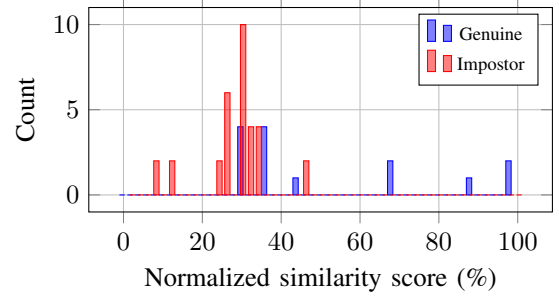


Figure 18: LivDet score histogram using the best score over probe templates.

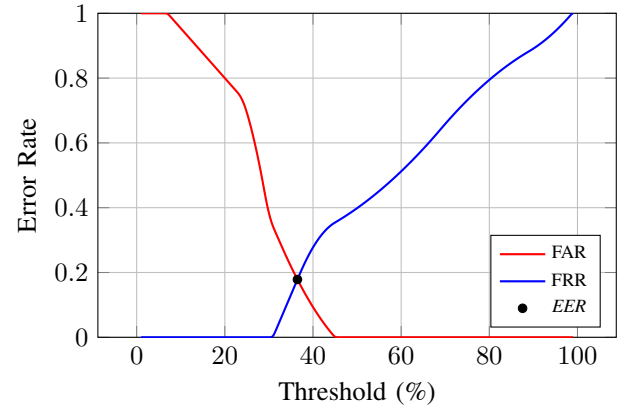
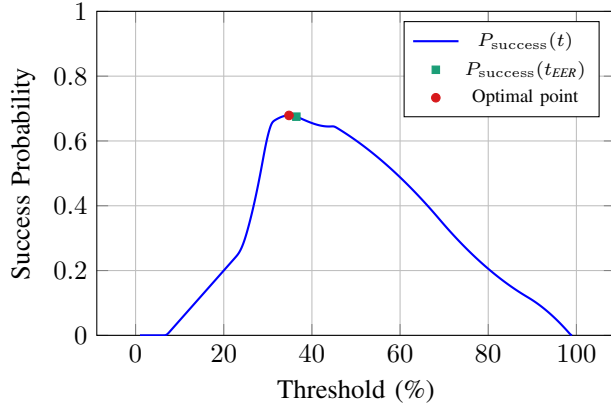


Figure 19: *FAR* and *FRR* as a function of the threshold using the best score over probe templates. The Equal Error Rate (*EER*) occurs at $\approx 17.8\%$.

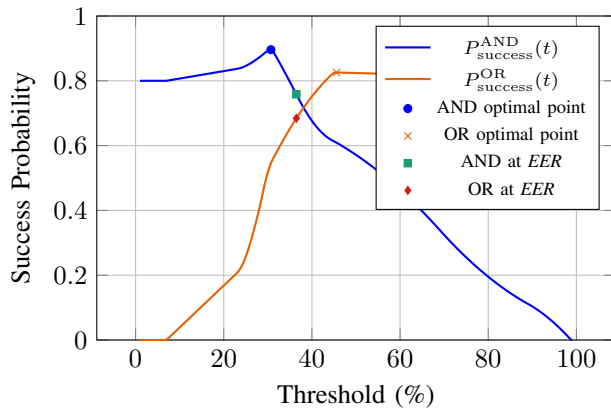
prevalant and easier on biometric systems in today's artificial intelligence era. Building new mechanisms, based on the suggested dynamic operating point done (§4) is a promising direction for future work. Our findings have important and impactful implications for the design of secure authentication biometric systems, as we suggest letting the integrators to set the thresholds of the biometric sensors. Harnessing our approach, amy affect millions of users around the world to enhance their security of digital and physical assets.

References

- [1] K. Abhishek, S. Roshan, P. Kumar, and R. Ranjan, "A comprehensive study on multifactor authentication schemes," in *Advances in Computing and Information Technology*, N. Meghanathan, D. Nagamalai, and N. Chaki, Eds. Berlin, Heidelberg: Springer Berlin Heidelberg, 2013, pp. 561–568.
- [2] Josh Koeber, Kalleigh Lane, "Password statistics: 82% of internet users practice dangerous password behaviors," All About Cookies, 2026. [Online]. Available: <https://allaboutcookies.org/password-users-behavior-survey>
- [3] A. Jain, L. Hong, and R. Bolle, "On-line fingerprint verification," *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 19, no. 4, pp. 302–314, 1997.
- [4] S. Prabhakar, S. Pankanti, and A. Jain, "Biometric recognition: security and privacy concerns," *IEEE Security & Privacy*, vol. 1, no. 2, pp. 33–42, 2003.



(a) Single biometric credential.



(b) AND and OR mechanisms.

Figure 20: Success probability as a function of the decision threshold using the best score over probe templates. The AND mechanism achieves a maximal success probability of 0.896, while the OR mechanism achieves 0.826. At the *EER* operating point, the success probabilities are 0.759 and 0.684 for the AND and OR mechanisms, respectively.

- [5] A. Sarkar and B. K. Singh, "A review on performance, security and various biometric template protection schemes for biometric authentication systems," *Multimedia Tools and Applications*, vol. 79, no. 37, pp. 27 721–27 776, 2020.
- [6] L. Hong and A. Jain, "Integrating faces and fingerprints for personal identification," *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 20, no. 12, pp. 1295–1307, 1998.
- [7] D. Maram, M. Kelkar, and I. Eyal, "Interactive multi-credential authentication," in *Proceedings of the 2024 on ACM SIGSAC Conference on Computer and Communications Security*, ser. CCS '24. New York, NY, USA: Association for Computing Machinery, 2024, p. 408–422. [Online]. Available: <https://doi.org/10.1145/3658644.3670378>
- [8] M. Moullem and I. Eyal, "Asynchronous authentication," in *Proceedings of the 2024 on ACM SIGSAC Conference on Computer and Communications Security*, ser. CCS '24. New York, NY, USA: Association for Computing Machinery, 2024, p. 3257–3271. [Online]. Available: <https://doi.org/10.1145/3658644.3670328>
- [9] I. Eyal, "On Cryptocurrency Wallet Design," in *3rd International Conference on Blockchain Economics, Security and Protocols* (*Tokenomics 2021*), ser. Open Access Series in Informatics (OASICS), V. Gramoli, H. Halaburda, and R. Pass, Eds., vol. 97. Dagstuhl, Germany: Schloss Dagstuhl – Leibniz-Zentrum für Informatik, 2022, pp. 4:1–4:16. [Online]. Available: <https://drops.dagstuhl.de/entities/document/10.4230/OASICS.Tokenomics.2021.4>
- [10] Shruti Kulkarni, Jeroen Kemperman. (2023) Beyond the password: Google workspace brings a major security innovation to customers with passkeys. Google Workspace. [Online]. Available: <https://workspace.google.com/blog/product-announcements/major-security-innovation-passkeys>
- [11] A. Mahfouz, R. Abdulghafor, and A. A. K. Ismaeel, "Passkeys in practice: An empirical evaluation of fido2/webauthn compliance and interoperability," in *2025 IEEE 24th International Conference on Trust, Security and Privacy in Computing and Communications (TrustCom)*, 2025, pp. 2781–2788.
- [12] S. Ribaric and I. Fratric, "A biometric identification system based on eigenpalm and eigenfinger features," *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 27, no. 11, pp. 1698–1709, 2005.
- [13] R. Vazan, "SourceAFIS: A Fingerprint Recognition Algorithm," <https://sourceafis.machinezoo.com>, 2018.
- [14] G. L. Marcialis, A. Lewicke, B. Tan, P. Coli, D. Grimberg, A. Congiu, A. Tidu, F. Roli, and S. Schuckers, "First international fingerprint liveness detection competition—livdet 2009," in *Image Analysis and Processing – ICIAP 2009*, P. Foggia, C. Sansone, and M. Vento, Eds. Berlin, Heidelberg: Springer Berlin Heidelberg, 2009, pp. 12–23.
- [15] D. Yambay, L. Ghiani, P. Denti, G. L. Marcialis, F. Roli, and S. Schuckers, "Livdet 2011 — fingerprint liveness detection competition 2011," in *2012 5th IAPR International Conference on Biometrics (ICB)*, 2012, pp. 208–215.
- [16] L. Ghiani, D. Yambay, V. Mura, S. Tocco, G. L. Marcialis, F. Roli, and S. Schuckers, "Livdet 2013 fingerprint liveness detection competition 2013," in *2013 International Conference on Biometrics (ICB)*, 2013, pp. 1–6.
- [17] V. Mura, L. Ghiani, G. L. Marcialis, F. Roli, D. A. Yambay, and S. A. Schuckers, "Livdet 2015 fingerprint liveness detection competition 2015," in *2015 IEEE 7th International Conference on Biometrics Theory, Applications and Systems (BTAS)*, 2015, pp. 1–6.
- [18] R. Morris and K. Thompson, "Password security: a case history," *Commun. ACM*, vol. 22, no. 11, p. 594–597, Nov. 1979. [Online]. Available: <https://doi.org/10.1145/359168.359172>
- [19] E. Reisinger, L. Genthner, J. Kerssemakers, P. Kensche, S. Borufka, A. Jugold, A. Kling, M. Prinz, I. Scholz, G. Zipprich, R. Eils, C. Lawrenz, and J. Eils, "Otp: An automatized system for managing and processing ngs data," *Journal of Biotechnology*, vol. 261, pp. 53–62, 2017, bioinformatics Solutions for Big Data Analysis in Life Sciences presented by the German Network for Bioinformatics Infrastructure. [Online]. Available: <https://www.sciencedirect.com/science/article/pii/S0168165617315924>
- [20] S. M. R. Security) and N. F. (HYPR), "Fido alliance white paper: Choosing fido authenticators for enterprise use cases," Technical Report, FIDO Alliance, 2022. [Online]. Available: <https://fidoalliance.org/wp-content/uploads/2021/09/FIDO-White-Paper-Choosing-FIDO-Authenticators-for-Enterprise-Use-Cases.pdf>
- [21] A. Jain, A. Ross, and S. Prabhakar, "An introduction to biometric recognition," *IEEE Transactions on Circuits and Systems for Video Technology*, vol. 14, no. 1, pp. 4–20, 2004.
- [22] M. Bishop, *Introduction to Computer Security*. Addison-Wesley Professional, 2004.
- [23] F. B. Schneider, "Authentication for people (draft textbook chapter)," <https://www.cs.cornell.edu/fbs/publications/chptr.AuthPeople.pdf>, 2009, retrieved October 2022.

- [24] R. Cappelli, D. Maio, D. Maltoni, J. Wayman, and A. Jain, "Performance evaluation of fingerprint verification systems," *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 28, no. 1, pp. 3–18, 2006.
- [25] D. Maio, D. Maltoni, R. Cappelli, J. Wayman, and A. Jain, "Fvc2000: fingerprint verification competition," *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 24, no. 3, pp. 402–412, 2002.
- [26] "Fingerprint verification competition 2002 (fvc2002)," <http://bias.csr.unibo.it/fvc2002/>, 2002, biometric Systems Laboratory, University of Bologna.
- [27] "Fingerprint verification competition 2004 (fvc2004)," <http://bias.csr.unibo.it/fvc2004/>, 2004, biometric Systems Laboratory, University of Bologna.

Appendix